CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable

communication.

Sol..

- Application → Presentation → Session → Transport → Network → Data Link → Physical

🧩 Missing Layers

1. Session Layer (between Presentation and Transport)
   - Function: Manages sessions (dialogues) between applications. It establishes, maintains, and terminates communication sessions.
   - Importance: Ensures that data streams are properly synchronized and organized, especially in long-running communications (e.g., video calls, file transfers). Without it, applications might lose track of ongoing conversations.
2. Data Link Layer (between Network and Physical)
   - Function: Provides reliable node-to-node data transfer. It handles error detection/correction, framing, and flow control.
   - Importance: Ensures that data sent over the physical medium (like cables or wireless signals) is free from transmission errors. Without it, raw bits could be corrupted or lost, making communication unreliable.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with

their respective OSI layers and analyze how transformation of data units supports

transmission.

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OSI Layer	Data Unit	Function in Transmission
Transport	Segment	Breaks application data into manageable pieces, adds sequence numbers and error-checking for reliable delivery.
Network	Packet	Encapsulates segments with logical addressing (IP addresses) to route data across networks.
Data Link	Frame	Adds physical (MAC) addressing and error detection for node-to-node delivery within the same network.
Physical	Bits	Converts frames into electrical, optical, or radio signals for transmission over the medium.

- **Segmentation (Transport Layer)**
  - Divides large data streams into smaller, ordered segments.
  - Ensures reliable delivery through acknowledgments and retransmissions.
- **Packetization (Network Layer)**
  - Adds source and destination IP addresses.
  - Enables routing across multiple networks, ensuring data reaches the correct endpoint.
- **Framing (Data Link Layer)**
  - Wraps packets with MAC addresses and error-check codes.
  - Guarantees that data moves correctly between devices on the same local network.
- **Bit Conversion (Physical Layer)**

- Translates frames into raw bits (0s and 1s).
- Sends them as electrical pulses, light waves, or radio signals through cables or air.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Sol..

OSI Layer	Address Type	Delivery Scope	Function
Data Link Layer	MAC Address	Local Delivery	Identifies devices within the same local network (LAN). Ensures frames reach the correct hardware interface.
Network Layer	IP Address	Global Delivery	Provides logical addressing for devices across different networks. Enables routing through multiple networks to reach distant destinations.

1. **Local Delivery (Data Link Layer)**
   - The **MAC address** is a unique hardware identifier assigned to each network interface card (NIC).
   - It ensures that data frames are delivered accurately **within the same network segment** — for example, from one computer to another connected to the same router or switch.
   - This layer handles **error detection and frame synchronization**, ensuring reliable node-to-node communication.
2. **Global Delivery (Network Layer)**
   - The **IP address** provides a logical identifier that can change depending on the network.
   - It enables **routing** — the process of finding the best path across interconnected networks (LANs, WANs, Internet).
   - Routers use IP addresses to forward packets from one network to another until they reach the

destination device.

3. **Combined Operation**
    - When data travels across networks, it moves from **global addressing (IP)** to **local addressing (MAC)** at each hop.
    - Routers strip and rebuild frames with new MAC addresses while keeping the IP address constant, ensuring that data moves seamlessly from source to destination.
    - This layered cooperation guarantees **end-to-end delivery** — from the sender's application down to the receiver's hardware.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data

transmission.

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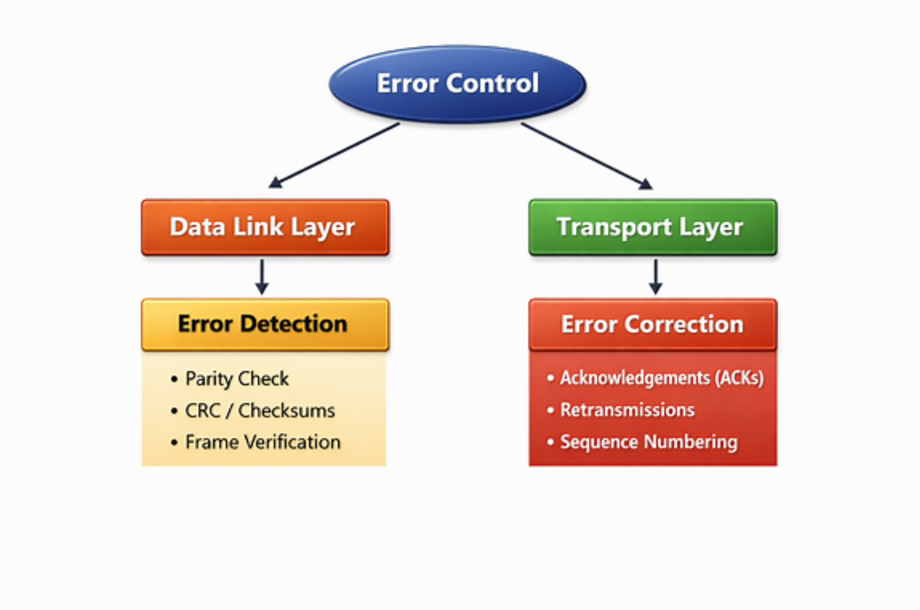

- **Data Link Layer → Error Detection**
    - Uses mechanisms like checksums, CRC (Cyclic Redundancy Check), and parity bits.

- Detects errors that occur during transmission across a single physical link (e.g., corrupted frames due to noise or interference).
- If an error is detected, the frame may be discarded or a retransmission requested.

- **Transport Layer → Error Correction**
  - Ensures end-to-end reliability across the entire communication path.
  - Uses techniques like acknowledgments (ACKs), retransmissions, and sequence numbering.
  - Corrects errors by resending lost or corrupted segments, ensuring complete and ordered delivery to the application.

5. Develop a concept map for web browsing using OSI layers and analyze how

failure in one layer affects the entire communication process.

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OSI Layer	Role in Web Browsing	Example Protocols	Failure Impact
Application	Provides the interface for users and web services (e.g., browsers, HTTP requests).	HTTP, HTTPS	Browser cannot display web pages or communicate with servers.
Presentation	Translates data formats (HTML, JPEG, encryption/decryption).	SSL/TLS	Data becomes unreadable or insecure.
Session	Manages sessions between browser and server (login persistence, cookies).	NetBIOS, RPC	Connection drops or session resets unexpectedly.
Transport	Ensures reliable data transfer between client and server.	TCP, UDP	Data loss, incomplete page loads, or broken connections.
Network	Routes packets across networks using logical addressing.	IP	Website unreachable due to routing failure.
Data Link	Handles local delivery and error detection on the same network.	Ethernet, Wi-Fi	Frames lost or corrupted; local connection unstable.
Physical	Transmits raw bits through cables or wireless signals.	Fiber, Copper, Radio	No connectivity — browser cannot send or receive data.

- **Top-Down Dependency**
  - Each layer depends on the one below it. For example, HTTP (Application Layer) relies on TCP (Transport Layer) to deliver data reliably.

- **Failure Cascade**
  - **A failure at any layer breaks the chain:**
    - If the Physical Layer fails (e.g., cable unplugged), all higher layers lose connectivity.
    - If the Network Layer fails (e.g., IP misconfiguration), routing stops, and the browser cannot reach the server.
    - If the Transport Layer fails, packets may be lost, causing incomplete page loads.
    - If the Application Layer fails, even though data arrives, the browser cannot interpret or display it.
- **Reliability Through Cooperation**
  - The layers work together to ensure that web browsing is seamless — from electrical signals to readable web content.
  - Each layer adds its own form of control (error checking, addressing, encryption) to maintain integrity and security.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

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Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

This means:

- **Physical Layer:** ✅ Working — cables, signals, and hardware are functioning.
- **Data Link Layer:** ✅ Working — local communication (MAC addressing, frame delivery) is successful.
- **Network Layer:** ❌ Failing — routing or IP addressing issues prevent packets from reaching other networks.
- **Transport Layer:** ❌ Failing — since the network layer is down, segments cannot be delivered end-to-end.

- **Application Layer:** ❌ Failing — web pages, emails, or apps cannot communicate because lower layers are broken.